Expert in Teams (EXT, F25)

University of Southern Denmark

Project Formulation

Machines in Motion Lab

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Abstract

In this Expert in Teams project, the Machines in Motion Lab group will design and create a miniature town model using LEGO with an automated train railway system. This town will feature multiple stops and multiple trains that will autonomously navigate the track from one station to the next. Management system software will be used to manage the track switches, lights, and trains using different controllers. The trains will be managed by a central system and could be expanded to make basic independent decisions like crash prevention. Additionally, the project will follow all the good development practices with requirements made in a MoSCoW format, milestones defined, and risk assessment done. As this project will be utilized as a marketing project, it will need to handle constraints on size, modularity, and resilience, which the team will research, document, and implement.

Contents

Abstract **1**
1 Introduction **1**
1.1 Context and Problem Background 1
1.2 Motivation and Relevance 2
2 Problem Formulation **3**
2.1 Problem statement 3
2.2 Scope of the project 3
2.2.1. Included in the scope 3
2.2.2. Excluded from the scope 3
2.3 Expectations and challenges 3
3 State of the art **5**
3.1 Previous SDU projects 5
3.2 Existing LEGO railway automation 5
4 Objectives and Requirements **6**
4.1 Objectives 6
4.2 Requirements 6
4.2.1 Functional requirements 7
4.2.2 Non-functional requirements 8
5 Choice of Technology, Tools, and Methods **9**
5.1 Technologies and Tools 9
5.1.1 Hardware 9
5.1.2 Software 9
5.2 Methods 10
6 Task Distribution **11**
6.1 Distribution table 11
6.2 Cross-disciplinary collaboration 12
7 Work Plan **13**
7.1 Explanation of each phase 13
8 Budget Overview **14**
8.1 Budget Table 14
9 Potential Risks and Mitigation **15**
9.1 Potential Risks 15
10 Bibliography **19**
Appendix A: Morphology Chart **20**
Appendix B: Gantt Chart **21**

1 Introduction

The University of Southern Denmark (SDU) Sønderborg offers several engineering programmes and this project is meant to create a miniature LEGO town model with train automation that can be transported, quickly assembled, and used by SDU representatives in an engaging manner at various events and exhibitions. The project is intended to catch the attention of potential new students and give them an opportunity to engage with the model while learning more about SDU Sønderborg and its engineering courses. The first event this project is intended for is in December.

1.1 Context and Problem Background

The project is carried out in collaboration with the Centre for Industrial Software [1] , one of the centres at the University of Southern Denmark. The project addresses the need for SDU Sønderborg to have a project that can be used to represent the university and its engineering programs at different events.

Research suggests that using innovative activities and engagement can help create and maintain interest in engineering [2]. This supports the idea of using a physical model to present the different engineering courses at SDU, instead of relying on standard advertisement materials, such as leaflets.

With this project, SDU would have a ready-made exhibit that can be used at events and is both engaging for the audience and easy to transport. This would reduce the need to create or borrow suitable projects for each event. This need is based partly on previous experience. In 2025, students represented the university at the 9th Silesian Science Festival (ŚFN) [7] and planned to use their semester project to represent the university and present their project to a broader audience. However, the project was not finished in time; thus, another project had to be prepared only a couple of weeks before the event. The available project was partially disassembled and not suitable for transport in a suitcase, making it inconvenient and stressful to prepare for the event. As a result, the project operated properly only for about half of the event.

Having a project specifically designed for use at events would also allow SDU to participate in more events without having to procure a new project each time, while still allowing expansion by new automation scenarios, increase of track layout, or addition of new input/output devices. One example is science or educational fairs such as the next edition of the Silesian Science Festival (ŚFN) [8] in Katowice, where the project can be used to attract potential new students and give them an opportunity to learn more about the engineering programmes offered at the University of Southern Denmark in Sønderborg.

Lastly, the project could be used in different present or future research projects conducted at the SDU university, like checking engagement and results of different automation scenarios; measuring differences of sensor readings to practically deduce the most reliable or cost-effective ones for moving devices; figuring out the effectiveness of different control algorithms or automation software.

1.2 Motivation and Relevance

It is important for SDU to have a project that can be entertaining and impressive for different types of audiences, including engineers, hobbyists, and people with no engineering background. This allows SDU to promote its engineering programs with more than words and leaflets but also through an actual project made by students at SDU. The project can also contribute to promoting science and engineering to a broader audience. By allowing people to interact with the model, it can give them a better understanding of the technologies and work involved in engineering. This can help create interest in engineering education, whether at SDU or elsewhere. The project also gives the students involved experience with completing a project from start to finish based on real requirements and constraints. This allows knowledge from their studies to be applied to a project with a practical purpose outside the university.

2 Problem Formulation

This section will define the problem faced, specify what is and is not included in the scope of the project, and finally provide an explanation and possible challenges of the solution.

2.1 Problem statement

"How can the University of Southern Denmark in Sønderborg advertise its courses effectively in Denmark and abroad at events and science fairs by the use of a relevant demonstrator, which would lead to an increase in the number of applicants to English courses at the Sønderborg campus?"

2.2 Scope of the project

The project is naturally limited by the resources and time provided, which are further elaborated in a subsequent Chapter 4. The scope will be split between two parts: Included and Excluded, which will be stated in the following sections.

2.2.1. Included in the scope

- Building and designing the track based mostly on LEGO bricks, with the potential of 3D printing specific parts.
- Building and designing a miniature model town based on business case and track layout constraints
- Adding peripheral output devices like LED lamps or switches.
- Adding peripheral input devices, for example, NFC readers, Reed sensors, Hall effect sensors, or other sensors to detect changes.
- Research and testing of different sensor types to find out the most reliable and cost-effective solution.
- Transport considerations - modular design with more layout controllers.
- Wired and wireless communication between devices via specified protocols.
- Train management system (TMS) integration and configuration.
- Building the trains from LEGO sets with small adjustments for the needed integrations.
- Useful visualization of the track layout and trains in them.
- Dashboard for real-time supervising the movement of the trains and demonstration purposes.
- Design and implementation of automation scenarios for expected use cases.

2.2.2. Excluded from the scope

- Designing the rails from the ground up.
- Use of light sensors.
- Designing the trains from the ground up and using custom designs.
- Fully self-made automation system.
- Support for seamless track changes.
- Use of Bluetooth for communication between TMS and microcontrollers.

2.3 Expectations and challenges

One of the most crucial tasks of the project is the wireless-connected communication that will happen between the Management System and the controllers. Software used for that will be the decision maker for the trains, while the controllers will operate the physical components on the designed railway. So, two-way communication is necessary for the system to function properly. Controllers will give information such as the location of the trains and their directions, and based on these inputs, the central management system will give commands to the controllers to act out on the trains, switches, signals, and crossings. Another difficulty, which is also an important piece of the working principle of the project, is the train detection system, which provides key information in order for the management system to work accordingly. The possibility of making a working charging station incorporated into the system is a task that could pose both mechanical and electrical challenges.

3 State of the art

To consider existing solutions, this chapter will first examine what SDU has previously presented at similar events. This will be followed by an analysis of existing LEGO and model railway automation systems and how they address the identified needs.

3.1 Previous SDU projects

The most similar project to be referenced is the earlier ŚFN 2025 project. The project featured a crane model designed and constructed by mechanical engineering students. This was a good representation of the mechanical and mechatronic programmes, however, it did not fully combine the content of all the bachelor programmes offered in Sønderborg, which is what this year's project strives for.

3.2 Existing LEGO railway automation

Model train automation, including LEGO set trains [12], is a well-established hobby among fans and enthusiasts, with numerous existing solutions. One such relevant example is the MattzoBricks [3] Train Automation System. Unlike many of the existing solutions that offer ready-made hardware and software, MattzoBricks also documents the design choices behind its system, which have heavily influenced the project's design and the automation software choices. Other similar sources include Tribrix [9], which handles custom LEGO parts compatible with LEGO trains and standardized sets, and 4dbrix [10], with simple automation built on Arduinos [11].

Another existing software solution is the Rocrail management train management system [4], which is a proprietary software application for model railway automation; it provides a network-based communication through TCP/IP and allows integration of many input and output devices. Although Rocrail is designed for model railways in general rather than specifically LEGO railways, its use in the MattzoBricks [3] system demonstrates its compatibility with LEGO-based railway automation. There is an open-source solution built specifically for the standard LEGO trains and the built-in PoweredUp ecosystem [13], called BrickRail [14], yet it makes handling of custom devices harder.

4 Objectives and Requirements

This chapter defines clear objectives and requirements, which ensure that the system design decisions remain aligned with high-level goals while providing verifiable benchmarks for final testing and validation.

4.1 Objectives

1. **Primary objectives**: core aims the project must achieve.
   - Build an autonomous railway system.
   - Manage communication between the track and trains using a management system.
   - Build the communication between the management system and controllers via Wi-Fi.
   - Design a unique, modular, and presentable miniature town using LEGO.
   - Build controllers for the trains, switches, and lights.
   - Create a train detection system using reliable sensors.
   - Design the rail track to support at least 2 trains simultaneously.
2. **Secondary objectives**: additional aims if time/resources allow.
   - Expand the track and system to support more than 2 trains.
   - Add a charging station to the system, ensuring reusability of the trains.
   - Add camera feedback from the train.
   - Add crash-preventing sensors and integration for smart driving.
   - Make a custom visualization layer for the tracks and state of the layout.
   - Ensure user-friendly customizability of the track layout and devices via smart documentation and software design.

4.2 Requirements

The functional and non-functional requirements are presented in the tables in their separate subsections. Main functional requirements were chosen by the team from the Morphology chart (Appendix A). They follow the MoSCoW prioritization method [5], which categorizes system requirements into four levels: ‘Must have’, ‘Should have’, ‘Could have’, and ‘Won’t have’. This helps define which features are essential for the system to function and which requirements are desirable but less critical.

4.2.1 Functional requirements

Functional requirements are a list of specific features and functions that the system must perform to satisfy project needs and business objectives. All functional requirements can be seen in Table 1 below.

Requirement	Description	Priority (MoSCoW)
F-1	The trains use batteries.	Must
F-2	The trains use a microcontroller.	Must
F-3	The trains communicate over WiFi.	Must
F-4	The trains and controllers communicate by the use of a TCP and predefined protocol.	Must
F-5	Have a reliable sensing technology for trains.	Must
F-6	Switches can change the track based on a signal via a servo motor.	Must
F-7	Specialized software will be used for creating the layout management system.	Must
F-8	Have a dedicated LAN network for the train connectivity.	Must
F-9	It is possible to monitor the trains' battery level.	Must
F-10	The trains have a physical on/off button.	Must
F-11	Each layout controller has a separate turn-on/off switch.	Must
F-12	The trains can be charged.	Should
F-13	The trains have an IMU sensor.	Should
F-14	LEDs can be controlled from the management system	Should
F-15	A section of the track is designated for charging the trains.	Should
F-16	Have a redundancy mechanism (connectivity & power supply) for the track controllers.	Should
F-17	The whole layout has an on/off switch.	Should
F-18	The trains can detect obstacles via a dedicated sensor.	Could
F-19	Trains can get controlled via Lego Bluetooth pad.	Could
F-20	Have barriers controlled by servos for sections/crossings.	Could
F-21	Have a display for the train timetable.	Could
F-22	The management system accounts for the overridden train movements (BLE, crash detection).	Could

Table 1: All functional requirements following the MoSCoW framework

4.2.2 Non-functional requirements

Non-functional requirements specify the way the system needs to be architected and its properties to satisfy project needs and business objectives. All non-functional requirements can be seen in Table 2 below.

Requirement	Description	Priority (MoSCoW)
NF-1	The town supports at least 2 train stations.	Must
NF-2	The track layout will be modular, allowing for seamless assembly and disassembly.	Must
NF-3	A management system must be easily deployable with layout already configured.	Must
NF-4	The entire setup must be easily packed into up to 4 suitcases.	Must
NF-5	The trains and layout are reliable.	Must
NF-6	The project fits in the budget.	Must
NF-7	The layout has a structured wiring management system.	Must
NF-8	The railway system should fit into the borders of an 180:160 cm rectangle.	Must
NF-9	Have a private version control system for the project files.	Must
NF-10	The town has an SDU-inspired layout.	Should
NF-11	The project is cost-effective.	Should
NF-12	The project needs to look marketable and professional.	Should
NF-13	The wireless connection is secured.	Could
NF-14	The connection will be redundant by connecting the layout controllers together.	Could
NF-15	The dashboard will be user-friendly.	Could

Table 2: All non-functional requirements following the MoSCoW framework

5 Choice of Technology, Tools, and Methods

In order to accomplish the goals and objectives of the project, several technologies and tools will be utilized in the domains of software and hardware design. Further simulations will be executed to verify design work and its concepts. Therefore, structural methodologies will be used to build a system architecture that ensures efficient execution and that the requirements are achieved.

5.1 Technologies and Tools

The project's technical components and tools will be described with the split between hardware and software.

5.1.1 Hardware

- **Microcontroller:** Microcontrollers will be used as a tool controlled by the management system to manipulate the states of the trains, switches, and lights. There are different microcontrollers that can be used, but one of the most suitable for the functional requirements is the ESP32. It has multiple wireless connectivity options, strong processing power, a small footprint, and is easily integratable into the management system.
- **Electronics CAD:** For the design of the electrical schematics of the project, it is necessary to use a software tool for designing and documenting connections between the different electrical components of the train system. Designing of custom PCBs is also done in such a software tool. Both of these tasks can be done in the free KiCAD program, which has been previously used by some of the team members [6].
- **Simulations:** In order to prove the electrical and mechanical designs, simulation tools will be used in accordance with the needs of the designed part of the project. Some of the software tools that can be used for electronics simulations are LTspice, PLECS, and Matlab Simulink [15][16][17]. For mechanical designs, BlueBrick will be used to design and plan the LEGO track layout digitally before assembling the physical setup [18].
- **Mechanical CAD:** Siemens NX will be used for the mechanical CAD design of the project [19]. It will be used to design the buildings and other custom elements of the LEGO-based town, taking into consideration the dimensions of the LEGO system and the track layout.
- **3D printing:** 3D printing will be used to manufacture selected custom components for the LEGO-based town. The components will be designed in Siemens NX and prepared for printing using Bambu Studio [20]. This allows the team to create parts that are not available as standard LEGO components and adapt them to the requirements of the designed layout.

5.1.2 Software

- **Train automation software:** will be used for controlling the trains automatically. It needs to be able to work with LEGO-based trains. An example of that software is Rocrail [3].
- **TCP/IP Communication:** This type will be utilized with the appropriate protocols (like the Rocrail-based XML protocol) to communicate, as it is the simplest and also a standardized, reliable way to connect input and output devices to the train automations.
- **Git:** Git with the GitHub repository will be used to store different project files, including all the code and hardware designs as well as documents, in a single spot, while allowing for version control and progress tracking [21].

5.2 Methods

When it comes to the process of project delivery, several choices were made to ensure that deadlines are met and the project is satisfactory. The choices were made following the Agile methodology [22] and Scrum framework [23], yet edited for the project specification, scope, and constraints, which combine both hardware and software. Except for the technical overview, like requirements and risk assessment, process-wise the team has decided to:

- Meet on Mondays with the project supervisor, and if one cannot participate, read information on the meeting afterwards.
- Communicate transparently on Discord continuously.
- Work in subteams and update the team continuously.
- Be responsible for your own work and own it.
- Follow the tasks and deadlines.
- The team has made a workplan that explains the details on milestones and work between those, which can be read in subsequent Chapter 7.
- Rafal was chosen as the team lead, with the task of ensuring continuous development and transparent communication.

6 Task Distribution

The following tables (Table 3 and Table 4) will define the roles and responsibilities of each team member, as well as specify the expected ways of collaboration between the different domains.

6.1 Distribution table

Discipline	Responsibilities
Mechanical	Design and model the train tracks.
Software	Setup the projects and repositories.
Software	Configure the layout in the management system.
Mechanical	Design a modular and reliable town design that fits the train tracks layout.
Innovation & Business	Budget the project.
Innovation & Business	Create scenarios and marketing cases for the project.
Electronics & Mechatronics	Design controller PCBs for each train
Mechatronics	Hardware integration and testing
Mechatronics	Component layout on the track and their connections
Electronics & Mechatronics	Design track controller PCBs in order for the management system to control the whole automation system.
Software	Combine the data from multiple sensors to increase the accuracy by fusing several sensors
Electronics	Design a Battery Management System (BMS) / Power Supply System (PSS) with battery status monitoring for both the trains and the track controllers
Electronics & Mechatronics	Design the train detection system using sensors and the track controllers
Mechatronics & Software	Define the protocol in which communication must be done with the management system
Mechanical	3D printing necessary parts for the tracks and trains
Electronics & Mechanical	Managing of wires and cables

Table 3: An overview of each team member, the discipline they represent, and their responsibility

6.2 Cross-disciplinary collaboration

Discipline	Collaboration Responsibilities
Mechanical	• With Electronics - Wiring and component placement in the track design • With Mechatronics - Track layout • With Engineering Innovation & Business - Track layout organization
Mechatronics	• With Electronics - PCB Soldering • With Electronics and Mechanical - Wiring and component connections • With Software - Sensor fusion • With Mechanical - Track layout • With Electronics and Software - programming of microcontrollers
Software	• With Mechatronics- Will help with communication between the management system and the track layout. • With Electronics - Will help with monitoring the battery levels.
Electronics	• With Mechanical - Track layout design due to electronics placement and wiring • With Mechatronics - Track and train controller design • With Engineering Innovation & Business - Providing required components for a BOM file and PCB ordering
Business	• With Software: Design the marketable scenarios for the trains to follow • With Mechanical: Design the layout of the elements • With Mechatronics and Electronics: Cost of the components

Table 4: An overview of what responsibilities will be in collaboration with multiple disciplines

7 Work Plan

In order to deliver the project by the correct deadlines, a work plan has been decided; detailed dates can be seen in a Gantt chart in Appendix B, yet as a baseline the project has been split into 4 parts: Design, MVP, Polish and Report.

7.1 Explanation of each phase

Design is the first phase, where the group gets a shared perspective of what will be created. This is the phase when requirements, technologies, and objectives are defined. It is concluded with the Problem Formulation deliverable.

In the second phase - MVP (Minimal Viable Product) - the group will fulfill the Must requirements, making a fully functional product. It is finished by the middle of November, and the whole work takes an iterative process with weekly minor iterations, with major iterations taking 2 weeks. The difference being that major iterations will try to conclude with a functional version of the worked parts, while minor iterations may finish without finalized parts. This is taken to give the proper amount of time while keeping reflections continuous. This phase concludes with the working product and its documentation.

The Polish phase (third phase) focuses on handling technical debt and preparing the project for travel; it focuses on esthetic, documentational, and non-function-breaking tasks.

The final phase (Report) will overlap with other phases, as the report is going to be written iteratively over the project development, with most of the time being spent on it after the ŚFN and third phase from the 7th of December. This phase concludes with a finished report and project delivered through DigitalEksamen.

8 Budget Overview

The current cost of the identified components is estimated at 6 864.27 DKK. The budget includes the LEGO railway sets acquired before the preparation of the budget to provide a more realistic representation of the project cost. Standard LEGO trains and tracks were selected to reduce potential compatibility and construction problems. The trains will be modified rather than designed entirely from scratch. Ready-made microcontrollers, motor drivers, breadboards, cables, and other prototyping components will be used to develop and test the train control and communication systems before producing the final solution.

However, the budget is not yet complete, as additional sensors, PCB components, power supplies, recharging equipment, and soldering or crimping materials may be required. The budget will therefore be updated once the team finalizes the designs and confirms which components are already available.

8.1 Budget Table

Table 5 below shows the budget, as it has been estimated.

Component	Quantity	Unit Price (DKK)	Total Cost (DKK)
LEGO railway sets and components	1	3,400.00	3,400.00
ESP32	4	73.00	292.00
Servo motor	3	40.00	120.00
Breadboard Kit Jumper Wires Set	1	97.00	97.00
DIY Kit Jumper Wires	2	157.00	314.00
400 Tie-points Breadboard	3	46.99	140.97
LEGO PF cable/connector	1	139.00	139.00
9V DC Power Supply	1	99.00	99.00
USB cable	1	20.00	20.00
Assorted LED diodes	1	79.00	79.00
Lithium batteries	5	110.66	553.30
Hall Effect Sensor	3	200.00	600.00
Router	1	250.00	250.00
PCB	5	32.00	160.00
Micro-Servos	10	30.00	300.00
PLA filament	3	100.00	300.00
Current estimated cost			**6 864.27**

Table 5: The estimated budget of the project

9 Potential Risks and Mitigation

During the development of the project, several potential risks may affect the successful completion of the project. The main risks are related to technical development and system integration, project organization, availability of resources, and time constraints. Risk mitigation involves the development of plans to manage, prevent, or reduce potential risks to an acceptable level. It will create preventive actions to reduce the probability of occurrence or the impact severity. The following table identifies the main project risks and the corresponding mitigation strategies.

9.1 Potential Risks

Potential risks have been explained in Table 6 below, specifying the probability of the risk happening (1 being very unlikely, 5 being certain) and their potential impact on the project (1 being insignificant impact, 5 being severe impact) with a combined field PxI, used to determine how severe a risk is. All of the risks were explained, and mitigation strategies have been supplied.

Technical Issues						
Risk ID	**Potential Risk**	**Description**	**Probability**	**Impact**	**PxI**	**Mitigation Strategy**
R-01	Communication failure between the management system and controllers	Loss or instability of communication may prevent commands and train data from being exchanged correctly.	4	5	20	Test communication early and regularly, implement reconnection mechanisms, and the manual system shutdown by adding an emergency button where possible.
R-02	Train detection or sensor failure	Incorrect or missing train detection may provide the management system with inaccurate information about train positions.	3	4	12	Test sensors under different operating conditions and verify train detection before allowing autonomous operation.
R-03	Software and system integration issues	Software bugs or incompatibility between the management system, controllers, sensors, and actuators may prevent the complete system from	3	5	15	Develop and test subsystems independently before integration and perform incremental integration testing.

		operating correctly.				
R-04	Faulty Wiring	Incomplete or incorrect wiring configurations can lead to components getting damaged or system failure	2	4	8	Relevant circuits should initially be simulated, checked, and tested before implementing them into the system
R-05	Hardware or power failure	Failure of microcontrollers, batteries, servos, or other electronic components may interrupt or shut down the system operation.	3	4	12	Test components before integration, monitor battery levels, and keep critical spare components available.
Organizational Issues						
R-06	Delays between dependent tasks or subteams	A delay in one subsystem may prevent another team from continuing development or integration.	3	4	12	Define responsibilities and dependencies clearly, establish milestones, and continuously communicate progress between subteams.
R-07	Communication and coordination issues	Miscommunication or unclear responsibilities may result in duplicated work, missing tasks, or integration problems.	2	2	4	Maintain regular meetings, use shared documentation and GitHub, and clearly assign responsibility for individual tasks.
R-08	Team members being absent due to illness	A group member may miss workdays and be unable to finish their work that was assigned, due to illness	1	2	2	Should be assessed by the management and their work should be given to another member
Resource-related Issues						
R-09	Component availability and supplier delays	Required controllers, sensors, servos, PCB components, or other parts may not arrive in time for development and testing.	2	5	10	Identify and order critical components early and identify suitable alternative components or suppliers.
R-10	Budget overrun	Component purchases, replacement parts, or	2	4	8	Track project expenses and prioritize

		design changes may cause the project to exceed the available budget.				components required for Must requirements before implementing Should/Could features.
R-11	Damage or setup problems during transportation	The modular railway system may be damaged, lose connections, or require excessive setup time when transported to the science fair.	3	3	9	Use a modular and transportable design, secure sensitive components during transport, label connections, and test the complete setup after reassembly.

Table 6: All the risks that were identified for the project, their probability of happening, their impact on the project, and their mitigation strategies

For easier visualization and summarization, all risks are then put into a table (Table 7 below) using the Risk IDs and placed according to PxI calculations. The table will be color-coded in order to show how much attention each risk requires.

		Probability				
Impact		Very Unlikely(1)	Unlikely (2)	Moderate (3)	Likely (4)	Certain (5)
	Insignificant (1)					
	Minor (2)	R-08	R-07			
	Moderate (3)			R-11		
	Major (4)		R-04 R-10	R-02 R-05 R-06		
	Severe (5)	R-09		R-03	R-01	

Table 7: Visual representation of the severity of the risks identified for the project

Legend:

green cells- small, insignificant risk

yellow cells- medium risk

orange cells- significant risk

red cells- impactful, huge risk

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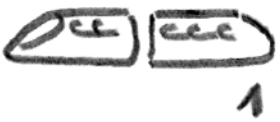
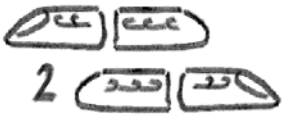
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Appendix A: Morphology Chart

Function	Solution 1	Solution 2	Solution 3
Number of trains			
Route selection	USER-SELECTED	PREDEFINED	AUTOMATIC ALGORITHM
Communication	WIRED CONNECTION	WI-FI	BLUETOOTH
Train control	ON-BOARD MICROCONTROLLER	PC-BASED	CENTRAL MICROCONTROLLER
Position detection	IR SENSORS	RFID SENSORS	SOFTWARE BASED
Track switching	LEGO MOTOR	SERVO MOTOR	MANUAL
User interface	WEB APP	MOBILE APP	PHYSICAL CONTROL PANEL

Appendix B: Gantt Chart

